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RESEARCH ARTICLE

Real-Time Spatiotemporal Gait Analysis Using a Hip-Assistive Wearable Robot

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ABSTRACT Wearable robots for lower-limb assistance have advanced significantly in recent years. However, unlike multi-joint assistive robots that support the hip, knee, or ankle joints, hip-only systems are typically equipped with fewer sensors, which limits their ability to support quantitative performance evaluation and gait analysis during use. This study presents a gait analysis algorithm embedded in a hip-assistive wearable robot (SUIT H10, Angel Robotics), designed to estimate key spatiotemporal and kinematic gait parameters using only onboard sensors. The algorithm performs real-time estimation of foot contact event, forward velocity, and knee joint angle using hip joint encoders and inertial sensors. Spatiotemporal gait parameters, including cadence, swing phase ratio, walking speed, and step/stride lengths, are subsequently computed through post-processing after short walking trials. The algorithm was validated through 64 walking trials involving 17 healthy adults, and its post-processed gait metrics were quantitatively compared with those from a commercial optical system (OptoGait). These results demonstrate the feasibility of quantitative gait assessment for hip-assistive robots without external measurement systems, supporting performance evaluation in both clinical and real-world settings.

INDEX TERMS Foot contact detection, gait analysis, kinematic modeling, real-time parameter estimation, sensor signal processing, wearable robot.

I. INTRODUCTION

Wearable robots have gained increasing attention as a means to improve gait function and mobility in both clinical and everyday environments. Early wearable robots were primarily designed for use on treadmills or in controlled environments because of their large size, limited sensing capabilities, and reliance on external power sources. Recent advances in hardware miniaturization, control algorithms, and sensing technology have enabled the development of lower-limb assistive devices capable of supporting over-ground walking in diverse user populations, including individuals with neurological impairments or age-related decline [1], [2]. To support safe and effective assistance, these systems require not only reliable control but also accurate methods to evaluate their performance during use.

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Although a wide range of control strategies for gait assistance has been proposed in recent years [3], [4], objective and standardized methods for assessing their effectiveness have not yet been established [5]. Traditional evaluation metrics such as metabolic cost offer a system-level view of energetic efficiency but are influenced by individual physiology, external conditions, and task variability [6], [7]. Electromyography (EMG) provides local neuromuscular information but suffers from noise sensitivity and complexity in interpretation [8]. These limitations have led to increasing interest in kinematic and spatiotemporal gait analysis as a more direct and interpretable means of evaluating wearable robot performance [9], [10]. For instance, improvement in step length or step time asymmetry enabled by robotic assistance in individuals with hemiplegia due to stroke can be provided as clinically meaningful measures [11].

To quantify such gait metrics, external motion capture systems, particularly marker-based optical setups as shown

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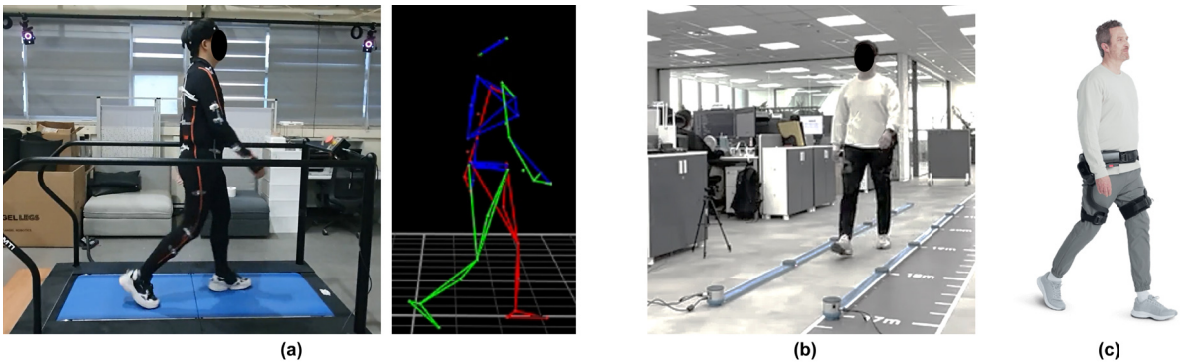

FIGURE 1. Illustration of gait analysis system configurations:(a) Conventional optical motion capture system using 3D markers and cameras;(b) System for optical detection made by a transmitting and a receiving bar for spatiotemporal gait parameter measurement;(c) Wearable robot system (SUIT H10) embedding the proposed gait analysis algorithm.

in Fig.1(a), have long been considered the gold standard in research environments due to their high spatial accuracy and full-body kinematic resolution [12]. However, these systems pose significant limitations when applied to experiments involving wearable robots. First, conventional optical motion capture systems rely on biomechanical models derived from unimpaired human movement. The addition of a wearable device alters joint dynamics, introduces mechanical coupling, and adds inertial loads, violating these model assumptions and leading to estimation errors. Second, reflective markers are ideally placed on anatomical landmarks (e.g., joint centers), but when an exoskeleton or soft robotic suit is worn, markers should often be attached to the surface of the device rather than the underlying limb. This deviation results in a mismatch between the modeled and actual limb kinematics, further degrading measurement fidelity. Third, because these systems require a fixed multi-camera setup within a calibrated capture volume, trials must be conducted on a treadmill rather than overground. This constraint introduces biomechanical discrepancies between treadmill and overground walking, limiting the ecological validity of the data [13].

To address these limitations, alternative motion analysis tools have been developed. Systems based on infrared walkway setups (e.g., OptoGait, Fig.1(b)) offer improved ease of use and portability [14], while wearable motion capture suits equipped with inertial measurement units (IMUs), such as the XSens MVN system [15], enable ambulatory data collection outside of the lab. Nonetheless, these systems remain constrained by key assumptions of human-only motion models and still require time-consuming calibration and post-processing procedures [16]. Furthermore, their performance may degrade in the presence of wearable robotic devices due to sensor occlusion, magnetic interference, or deviation from assumed limb dynamics.

In many hip-only assistive robots, gait phase estimation based on onboard joint encoders and inertial data has been commonly used to coordinate assistive torque [17]. However, these systems are generally not designed for detailed gait analysis or quantitative assessment of gait-related outcomes. Since only proximal joint sensors are available, important aspects of gait behavior such as the timing of foot contact or knee joint angle cannot be directly observed. As a result, motion analysis with hip-only systems has typically relied on external optical or pressure-based systems, which restrict applicability to laboratory settings and prevent evaluation in real-world environments.

Some multi-joint assistive robots have addressed this issue by incorporating onboard gait analysis capabilities that leverage sensors at the hip, knee, and ankle joints [18], [19]. While these systems demonstrate comprehensive sensing, applying similar approaches to hip-only devices remains challenging due to the absence of distal joint data.

Unlike these prior approaches, IMU-based gait estimation methods have predominantly relied on sensors attached to distal limb segments, such as the shanks or feet, to directly detect signals associated with ground contact [20]. Even in studies involving hip exoskeletons, which are mechanically similar to our system, additional IMUs mounted on the feet were deemed necessary to achieve robust locomotion mode recognition across varied terrains [21]. While effective, these multi-sensor configurations require users to wear multiple devices, which can be cumbersome and impractical for daily use. A recent review by Prasanth et al. [22] summarizes representative IMU-based approaches including threshold-based and state-machine methods, which primarily focus on detecting gait events such as heel strike and toe-off or recognizing locomotion modes. However, these methods typically do not estimate clinically relevant spatiotemporal metrics such as cadence or step length, which are essential for quantitative gait assessment.

To overcome these limitations, this paper presents a gait analysis algorithm embedded in a hip-assistive wearable robot (SUIT H10, Angel Robotics). The system estimates foot contact events, forward velocity, and knee joint angle in real time using only the hip joint encoder data and inertial signals. Subsequently, spatiotemporal gait parameters such as cadence, swing phase ratio, walking speed, and step/stride lengths are computed through post-processing

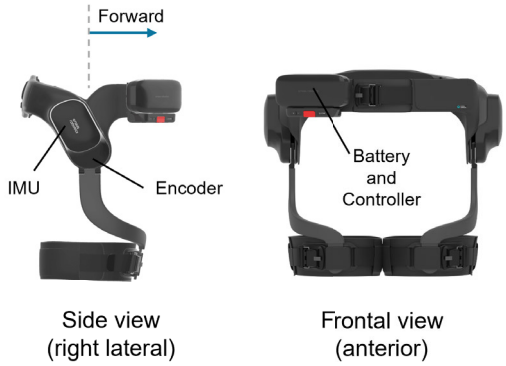

FIGURE 2. Side view showing sensor placement and frontal view illustrating the locations of the battery and processor. The forward walking direction is indicated by the arrow.

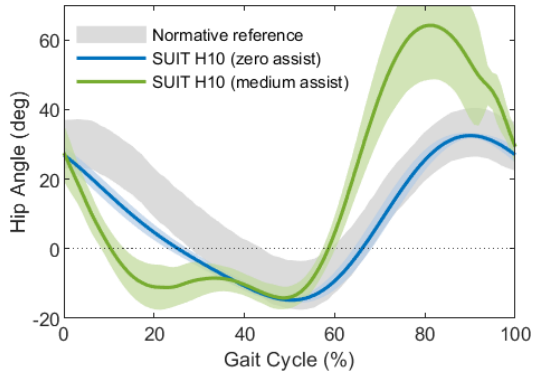

FIGURE 3. Hip joint angle trajectories (mean ± SD) with and without assistance, compared against normative data [10] The robot's embedded sensors reveal clear changes in gait pattern under assistance, which are difficult to capture using conventional motion analysis systems.

after short walking trials. The algorithm was validated against a commercial optical gait analysis system using data from healthy adults. This approach represents a significant step toward a more practical and user-friendly wearable system that enables quantitative gait evaluation without external measurement devices.

In contrast to prior IMU-based gait estimation methods that primarily focus on event detection or locomotion mode recognition, the present study emphasizes the estimation of clinically meaningful spatiotemporal parameters. These parameters are commonly used in clinical gait assessment and rehabilitation to support phase detection and the quantitative evaluation of gait performance.

II. WEARABLE MOTION ANALYSIS SYSTEM

A. SYSTEM CONFIGURATION

Hip-assistive wearable robots have shown strong potential for gait rehabilitation and mobility support, owing to the biomechanical importance of the hip joint in walking. The hip generates substantial joint torques during the stance phase, playing a key role in forward propulsion and postural stability [9]. Focusing assistance at the hip enables effective support with simpler mechanical design and lower control complexity compared to multi-joint systems, and clinical studies have demonstrated its practical benefits [6], [17].

SUIT H10, developed by Angel Robotics Co., Ltd., is a wearable device to provide gait assistance and basic motion analysis through embedded sensors placed at the hip. The device is worn around the waist and thighs, ensuring firm attachment and alignment with the wearer's hip joints. Motorized modules are positioned laterally to apply assistive torque directly to the hip joints during walking.

As a hip-only assistive robot designed for simplicity, SUIT H10 does not include sensors to measure the knee angles or ground contact (Fig. 2). Unlike conventional gait analysis systems that rely on optical markers or wearable foot sensors for motion tracking, SUIT H10 performs gait analysis using only onboard hip-mounted sensors, which are also used for gait assistance. This integrated configuration reduces the burden of sensor placement, improves user comfort, and enables more practical use in clinical or daily environments.

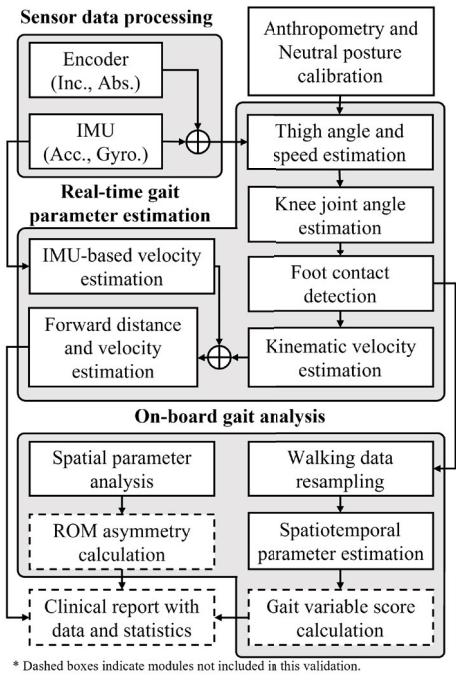

FIGURE 4. Flowchart of data processing from sensor input to gait analysis.

B. EMBEDDED GAIT ANALYSIS UNDER ROBOTIC ASSISTANCE

The embedded gait analysis system in SUIT H10 enables direct observation of the robot's assistive effects without reliance on external motion capture devices. Under the no-assist condition, the system captures hip joint kinematics that resemble known normative gait patterns, supporting its feasibility for gait analysis using only onboard sensors. As shown in Fig. 3, the mean hip angle trajectories without

assistance (blue solid line) closely follow normative data (gray shaded area) reported in previous studies [10].

More notably, when assistive torque is applied, the measured hip joint trajectories exhibit distinct changes compared to the no-assist case. These alterations, such as increased hip flexion during swing (green solid line in Fig. 3), clearly reflect the influence of robotic assistance. Such synchronized changes are not easily observable using conventional motion analysis tools. For example, optical systems like VICON require precise marker attachment, which can be challenging when the user is wearing a robot.

In contrast, the proposed embedded system directly captures these assistive effects from within the robot, providing a practical and precise solution for evaluating both pre- and post-assistance gait characteristics. This capability supports practical and repeatable evaluation of assistive effects using only the robot's onboard sensors, enabling performance assessment in both clinical and real-world settings.

C. SENSOR-BASED GAIT PARAMETER ESTIMATION

SUIT H10 integrates two 9-axis Inertial Measurement Units (IMUs) and rotary encoders mounted on each hip joint drive module to measure joint kinematics. No additional sensors are used solely for gait analysis; instead, the same sensors used for torque control are repurposed for motion monitoring. Sensor data are sampled at 1,000 Hz and processed in real time at the same rate, synchronized with the assistive control loop.

The real-time algorithm begins with anthropometric calibration and neutral posture alignment. Using encoder and IMU signals, the system estimates thigh angle, angular velocity, and knee joint angle based on dynamic modeling. Foot contact event is estimated based on leg length variations derived from thigh and knee joint angles. Forward walking speed is computed using a combination of foot contact state and IMU-derived velocity. These parameters serve as input for spatiotemporal analysis and joint kinematic profiling in the post-processing pipeline, as illustrated in the upper section of Fig. 4.

D. POST-PROCESSING AND GAIT METRIC ANALYSIS

After short walking trials, the embedded system performs onboard post-processing to compute spatiotemporal gait parameters. Foot contact events are used to segment individual gait cycles, and each cycle is time-normalized for comparison across steps and subjects.

From the segmented data, five key gait metrics are estimated: cadence, walking speed, step length, stride length, and swing phase ratio. These parameters are derived from estimated foot contact event, forward velocity, and hip joint kinematics. The post-processing pipeline is implemented entirely on-board, without requiring external sensors or offline computation, as illustrated in the lower portion of Fig. 4. As illustrated in Fig. 4, the system also includes modules for range of motion (ROM) asymmetry analysis and clinical report generation, which were not evaluated in the current validation.

Although the system does not include sensors on the feet or knees, these metrics provide a clinically relevant summary of gait characteristics using only hip-mounted sensors. The estimation methods are designed to support consistent analysis across trials without requiring external motion capture systems.

III. REAL-TIME AND EMBEDDED GAIT PARAMETER ESTIMATION

This section describes the real-time estimation algorithms implemented in the embedded system. The approach is based on a simplified 2D biomechanical model of the lower limbs, enabling real-time gait event detection, joint kinematic estimation, and per-step estimation of spatiotemporal parameters such as cadence and step length using only onboard sensors.

In this study, the human body is modeled using four rigid segments: left thigh, left shank, right thigh, and right shank. A two-dimensional sagittal plane analysis is conducted, with bilateral symmetry explicitly considered. This segmental structure and coordinate definition are illustrated in Fig. 5. The hip joint center, the origin of the global coordinate system, is defined as $\boldsymbol{p}_{\text{hip}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

For simplicity, the hip joint is modeled as a single point at the center of the pelvis, and lateral pelvic width is neglected. That is, we define the origin of the coordinate system as a shared hip joint position $\boldsymbol{p}_{\text{hip}} = \boldsymbol{0}$ for both left and right legs. This assumption is commonly used in planar gait models and allows symmetric segment definitions without introducing additional horizontal offsets due to pelvic geometry.

Each leg is modeled using a thigh and shank segment, assumed to be symmetric in structure. Segment lengths are denoted as l_{thigh} and l_{shank}. For notational compactness, let $s \in \{L, R\}$ indicate the left and right side, respectively. The knee and ankle joint positions defined as follows:

$$\boldsymbol{p}^s_{\text{knee}} = \begin{bmatrix} l_{\text{thigh}} \sin(\theta^s_{\text{thigh}}) \\ -l_{\text{thigh}} \cos(\theta^s_{\text{thigh}}) \end{bmatrix} \tag{1}$$

$$\boldsymbol{p}^s_{\text{ankle}} = \boldsymbol{p}^s_{\text{knee}} + \begin{bmatrix} l_{\text{shank}} \sin(\theta^s_{\text{shank}}) \\ -l_{\text{shank}} \cos(\theta^s_{\text{shank}}) \end{bmatrix} \tag{2}$$

The position of the shank's center of mass (COM) is given by:

$$\boldsymbol{p}^{\text{shank},s}_{\text{com}} = \boldsymbol{p}^s_{\text{knee}} + \alpha_{\text{shank}} \begin{bmatrix} l_{\text{shank}} \sin(\theta^s_{\text{shank}}) \\ -l_{\text{shank}} \cos(\theta^s_{\text{shank}}) \end{bmatrix} \tag{3}$$

Here, the segment orientation angles θ^s_{thigh} and θ^s_{shank} are defined as the angles measured counterclockwise from the vertical downward direction. That is, a perfectly upright segment aligned with the vertical axis has an angle of zero. The scalar α_{shank} denotes the relative location of the shank's center of mass along its length, measured from the knee joint. In this model, we set $\alpha_{\text{shank}} = 0.5$, assuming the center of mass is located at the midpoint of the segment.

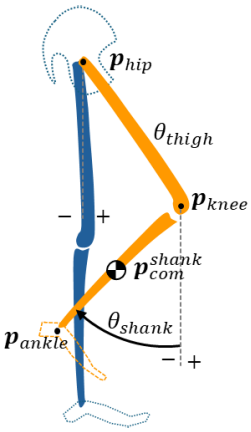

FIGURE 5. Schematic representation of the planar five-segment human body model used in this study. Each leg is composed of a thigh and a shank segment, symmetrically arranged about the pelvis. The hip joint center p_{hip} is defined as the origin of the coordinate system. Joint positions p_{knee} and p_{ankle}, as well as segment orientation angles θ_{thigh} and θ_{shank}, are illustrated for both left and right sides.

A. FOOT CONTACT DETECTION

Accurate detection of the swing and stance phases is important for gait analysis, as it influences the calculation of temporal gait parameters. Since foot-mounted sensors, such as pressure insoles or force plates, are excluded to simplify the system, this study proposes a method to infer foot contact events based on changes in leg length during walking.

For the construction of a two-dimensional kinematic model of the lower limbs, the subject's anthropometric measurements, including height, body weight, and thigh and shank lengths, should be provided. Then, the vertical distance from the hip joint to the ankle is estimated during the gait cycle, decreasing during the swing phase and returning to the standing posture during the stance phase.

As the SUIT H10 robot does not include a knee joint sensor, the knee angle is estimated to formalize the vector from the hip joint to the ankle joint for leg as $\boldsymbol{r}^s_{\text{hip}\to\text{ankle}}$ for both legs where $s \in \{L, R\}$. The vertical length of this vector, corresponding to the leg's projected extension along the vertical axis, is given by:

$$l^s_{\text{vertical}} = \left|\left[\boldsymbol{r}^s_{\text{hip}\to\text{ankle}}\right]_y\right| = \left|\left[\boldsymbol{p}^s_{\text{ankle}} - \boldsymbol{p}_{\text{hip}}\right]_y\right|. \tag{4}$$

During walking, the vertical hip-to-ankle length decreases in early swing due to knee flexion, then increases again in late swing as the knee extends, approaching a near-maximum value during stance. However, because some individuals maintain slight joint flexion even while standing (e.g., a light crouch posture), a short vertical length alone does not always indicate a swing phase.

Therefore, in this study, the difference in estimated vertical leg length between the two legs is used to infer the swing and double stance phases. This differential measure helps distinguish between true swing and symmetric flexed postures. To improve robustness against signal noise and oscillations near the threshold boundary, a hysteresis-based approach is applied using separate rising and lowering thresholds. Let γ_{rise} and γ_{fall} denote the rising and lowering threshold factors, respectively, such that $\gamma_{\text{fall}} < \gamma_{\text{rise}}$. In this study, we empirically set $\gamma_{\text{rise}} = 0.03$ and $\gamma_{\text{fall}} = 0.0172$, based on pilot tests to ensure stable detection across participants. We denote the foot contact state for leg $s \in \{L, R\}$ at time t as $c^s(t) \in \{0, 1\}$, where 1 indicates contact (stance phase) and 0 indicates swing.

The foot contact state for each leg is determined based on the vertical hip-to-ankle length difference between the two legs. Let the absolute vertical difference be defined as:

$$\Delta l(t) = \left|l^L_{\text{vertical}}(t) - l^R_{\text{vertical}}(t)\right|. \tag{5}$$

The contact states $c^L(t)$ and $c^R(t)$ are updated according to the following rules:

- **Case 1 (Double stance)**:
  If $\Delta l(t) < \gamma_{\text{fall}} \cdot l^{\max}_{\text{leg}}$, then both feet are in contact: $c^L(t) = 1, \quad c^R(t) = 1$
- **Case 2 (Left stance)**:
  If $\Delta l(t) > \gamma_{\text{rise}} \cdot l^{\max}_{\text{leg}}$ and $l^L_{\text{vertical}}(t) < l^R_{\text{vertical}}(t)$, then the left foot is in contact: $c^L(t) = 1, \quad c^R(t) = 0$
- **Case 3 (Right stance)**:
  If $\Delta l(t) > \gamma_{\text{rise}} \cdot l^{\max}_{\text{leg}}$ and $l^R_{\text{vertical}}(t) < l^L_{\text{vertical}}(t)$, then the right foot is in contact: $c^L(t) = 0, \quad c^R(t) = 1$
- **Case 4 (No change)**:
  Otherwise, the contact state remains unchanged: $c^L(t) = c^L(t-1), \quad c^R(t) = c^R(t-1)$

In this framework, the instantaneous leg length is computed based on the thigh angle (θ_{thigh}) and the shank angle (θ_{shank}), along with subject-specific anthropometric measurements (i.e., thigh and shank lengths). While θ_{thigh} is directly measured using the hip joint encoders and IMU, θ_{shank} is not available from any physical sensor in the system.

This necessitates the estimation of the shank angle, which serves as a key input to the leg posture reconstruction and foot contact detection algorithm. The following section describes the proposed method for estimating both the shank angle and the knee joint angle, which is computed as the relative angle $\theta_{\text{knee}} = \theta_{\text{shank}} - \theta_{\text{thigh}}$, using only the available sensor data.

B. TWO-DIMENSIONAL KINETIC ANALYSIS

1) QUASI-STATIC GROUND REACTION FORCE ESTIMATION

To estimate the shank angle θ_{shank}, we begin by modeling the external forces acting on the shank segment. Fig. 6 illustrates the sagittal-plane Free Body Diagram (FBD) used in this analysis. Two forces are considered in the diagram: the Ground Reaction Force (GRF), applied at the ankle joint during stance, and the internal force transmitted from the thigh segment through the knee joint.

The GRF is modeled as a horizontal frictional force during foot-ground contact, as determined by the foot contact detection algorithm in Section III-A:

$$f^x_{\text{GRF}} = -k_{\text{friction}} v^x_{\text{ankle}}, \tag{6}$$

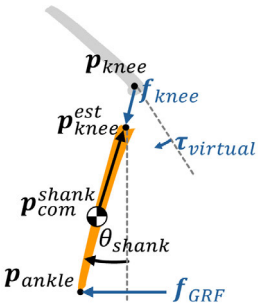

FIGURE 6. Schematic representation of the FBD.

where k_{friction} is an experimentally determined scaling factor, and v_{ankle}^{x} denotes the ankle's horizontal translational velocity. The internal force from the thigh, $\boldsymbol{f}_{\text{knee}}$, is modeled as a linear spring-damper system that tends to align the knee position with the expected position of the distal end of the shank:

$$\boldsymbol{f}_{\text{knee}} = k_{\text{spring}}\boldsymbol{e} + c_{\text{damping}}\dot{\boldsymbol{e}}. \tag{7}$$

Here, the error vector $\boldsymbol{e}$ is defined as

$$\boldsymbol{e} = \boldsymbol{p}_{\text{knee}} - \boldsymbol{p}_{\text{knee}}^{\text{est}}, \tag{8}$$

where $\boldsymbol{p}_{\text{knee}}$ is the observed knee joint position from thigh segment, and $\boldsymbol{p}_{\text{knee}}^{\text{est}}$ is the proximal estimated vector from the shank center of mass to the proximal end of the shank segment, expressed in the shank frame.

2) JOINT KINEMATICS ESTIMATION

Let $\boldsymbol{p}_{\text{com}}^{\text{shank}}$ denote the position of the shank's center of mass, and let m_{shank} represent the shank mass. Newton's second law yields that

$$m_{\text{shank}}\ddot{\boldsymbol{p}}_{\text{com}}^{\text{shank}} = \boldsymbol{f}_{\text{knee}} + \boldsymbol{f}_{\text{GRF}} + m_{\text{shank}}\boldsymbol{g}, \tag{9}$$

and the acceleration of the center of mass is approximated using finite differences over three consecutive time steps to implement this in real-time as:

$$\ddot{\boldsymbol{p}}_{\text{com}}^{\text{shank}}(k) \approx \frac{\boldsymbol{p}_{\text{com}}^{\text{shank}}(k+1) - 2\boldsymbol{p}_{\text{com}}^{\text{shank}}(k) + \boldsymbol{p}_{\text{com}}^{\text{shank}}(k-1)}{\Delta t^2}. \tag{10}$$

Here, Δt is the sampling interval, and $\boldsymbol{p}_{\text{com}}^{\text{shank}}(k)$ is inferred from joint kinematics under the assumption of rigid links. Rotational motion about the shank's center of mass is also considered. Let θ_{shank} denote the segment's orientation, and I_{shank} its moment of inertia about the COM. The torque balance equation is given by:

$$I_{\text{shank}}\ddot{\theta}_{\text{shank}} = r_{\text{shank}}^{\text{prox}} f_{\text{knee},\perp} - r_{\text{shank}}^{\text{distal}} f_{\text{GRF},\perp} + \tau_{\text{virtual}}, \tag{11}$$

where $r_{\text{shank}}^{\text{prox}}$ and $r_{\text{shank}}^{\text{distal}}$ are the moment arms of the applied forces relative to $\boldsymbol{p}_{\text{com}}^{\text{shank}}$, and $f_{\perp}$ denotes the perpendicular component of the respective force that contributes to torque. In the sagittal plane, these values reduce to scalars, and no cross products are required.

Here, τ_{virtual} is an auxiliary virtual torque applied to prevent unrealistic knee hyperextension. This term serves as a stabilizing constraint within the model and is not derived from any physical joint actuator or sensor. The angular acceleration is approximated using finite differences:

$$\ddot{\theta}_{\text{shank}}(k) \approx \frac{\theta_{\text{shank}}(k+1) - 2\theta_{\text{shank}}(k) + \theta_{\text{shank}}(k-1)}{\Delta t^2}. \tag{12}$$

The shank angle θ_{shank} is updated at each time step by solving the dynamics equations above, using observed kinematics and modeled external forces. The knee joint angle is then computed as the relative angle between the thigh and shank segments:

$$\theta_{\text{knee}} = \theta_{\text{shank}} - \theta_{\text{thigh}}. \tag{13}$$

This formulation enables the estimation of knee joint motion without direct sensing, relying only on pelvic IMU, hip joint encoder, and subject-specific limb lengths.

C. SPATIOTEMPORAL PARAMETER ESTIMATION

This subsection details the real-time implementation of key gait metrics introduced in Section II-D. Spatiotemporal parameters, including walking speed, step length, stride length, cadence, and stance/swing durations, are estimated using foot contact events identified in Section III-A. These metrics provide clinically relevant summaries of walking performance.

1) FORWARD VELOCITY AND DISTANCE ESTIMATION

The forward velocity is estimated by fusing IMU-based and kinematics-based velocities. The kinematic velocity v_{kin} is calculated from the horizontal displacement of the ankles:

$$v_{\text{kin}}(k) = \max\left(0, -\frac{\Delta \boldsymbol{p}_{\text{ankle}}^{x}(k)}{\Delta t}\right), \tag{14}$$

where the IMU-based velocity is derived from integrating global horizontal acceleration a_{L}^{x} and a_{R}^{x} as

$$v_{\text{IMU}}(k) = v_{\text{IMU}}(k-1) + \alpha_v \frac{a_{\text{L}}^{x}(k) + a_{\text{R}}^{x}(k)}{2}\Delta t, \tag{15}$$

and a fusion gain $\lambda_v \in [0, 1]$ is applied to combine the IMU and kinematics-based estimates:

$$v_x(k) = v_{\text{IMU}}(k) + \lambda_v \cdot (v_{\text{kin}}(k) - v_{\text{IMU}}(k)). \tag{16}$$

Here, α_v denotes the integration gain used for IMU acceleration, and $\lambda_v \in [0, 1]$ is a fusion coefficient that controls the contribution of the kinematics-based velocity estimate to correct IMU drift. These parameters were determined empirically based on pilot testing across multiple subjects and applied uniformly throughout the study. The cumulative walking distance is then updated as:

$$x_{\text{pos}}(k) = x_{\text{pos}}(k-1) + v_x(k)\Delta t. \tag{17}$$

TABLE 1. Baseline characteristics of participants (N=17).

Variable	Value	Min.-Max.	Q1	Q3
Sex (M:F)	9:8	-	-	-
Age (years)	31.6 ± 6.12	24-47	27	33
Height (m)	1.68 ± 0.07	1.51-1.80	1.63	1.71
Body weight (kg)	69.5 ± 14.1	45-97	58	80
Thigh length (m)	0.389 ± 0.036	0.350-0.475	0.365	0.39
Shank length (m)	0.435 ± 0.043	0.360-0.500	0.405	0.470

2) GAIT EVENT DETECTION AND STEP/STRIDE PARAMETERS

Heel strike and toe-off events are defined based on transitions in the binary contact state $c^s(t)$, where $s \in \{L, R\}$:

$$t_{\mathrm{HS},s} = \{t \mid c^s(t) = 1,\ c^s(t-1) = 0\}, \tag{18}$$

$$t_{\mathrm{TO},s} = \{t \mid c^s(t) = 0,\ c^s(t-1) = 1\}, \tag{19}$$

The forward position of the ankle for side s at time t is defined as:

$$x^s_{\mathrm{foot}}(t) = x_{\mathrm{pos}}[t] + p^x_{\mathrm{ankle},s}[t], \tag{20}$$

where $x_{\mathrm{pos}}[t]$ is the cumulative forward position of the pelvis, and $p^x_{\mathrm{ankle},s}[t]$ is the horizontal offset from the pelvis to the ankle. Using this definition, the step length is computed as:

$$l^{s,(n)}_{\mathrm{step}} = \left| x^s_{\mathrm{foot}}(t^{(n)}_{\mathrm{HS},s} - x^{\bar{s}}_{\mathrm{foot}}(t^{(n)}_{\mathrm{HS},\bar{s}}) \right|, \tag{21}$$

where $\bar{s}$ denotes the opposite side of s; that is, if $s = L$, then $\bar{s} = R$, and vice versa. Here, $t^{(n)}_{\mathrm{HS},s}$ denotes the time of the n-th heel strike on side s, and $n \in \mathbb{Z}^+$ indexes successive gait cycles.

The stride length is defined as the distance between two consecutive heel strikes on the same side:

$$l^{s,(n)}_{\mathrm{stride}} = \left| x^s_{\mathrm{foot}}(t^{(n+1)}_{\mathrm{HS},s}) - x^s_{\mathrm{foot}}(t^{(n)}_{\mathrm{HS},s}) \right|. \tag{22}$$

Cadence and phase durations are computed for each gait cycle n, using the timing of heel strike and toe-off events on side s. Specifically, cadence is defined as the number of steps per minute:

$$\mathrm{Cadence}^{(n)}_s = \frac{120}{t^{(n+1)}_{\mathrm{HS},s} - t^{(n)}_{\mathrm{HS},s}}, \tag{23}$$

and the durations of the stance and swing phases are calculated as:

$$T^{(n)}_{\mathrm{stance},s} = t^{(n)}_{\mathrm{TO},s} - t^{(n)}_{\mathrm{HS},s}, \tag{24}$$

$$T^{(n)}_{\mathrm{swing},s} = t^{(n+1)}_{\mathrm{HS},s} - t^{(n)}_{\mathrm{TO},s}. \tag{25}$$

IV. EXPERIMENTAL PROTOCOL AND VALIDATION

To evaluate the accuracy and clinical relevance of the proposed embedded gait analysis system, a series of walking trials were conducted using the SUIT H10 wearable robot. The experiment involved 17 healthy adult participants, all without known musculoskeletal or neurological disorders. All participants provided written informed consent prior to participation. The study involved minimal-risk testing with healthy adults and did not require formal Institutional Review Board approval under local institutional guidelines.

Participant demographics are summarized in Table 1. Anthropometric data, including body weight, height, and the segment lengths of the thigh and shank for both legs, were measured prior to testing. The thigh segment length l_{thigh} was measured as the linear distance from the greater trochanter (upper end of the femur near the hip) to the lateral femoral condyle (outer part of the knee joint), with the participant seated and the hip and knee flexed at approximately 90 °. The shank segment length l_{shank} was measured from the lateral femoral condyle to the lateral malleolus (outer ankle bone) under the same posture. All measurements were performed manually using a standard anthropometric tape or rigid ruler while ensuring consistent anatomical landmarks across participants.

A. EXPERIMENT PROTOCOL

Each participant was equipped with the SUIT H10 hip-assistive wearable robot and instructed to walk at a self-selected comfortable speed along a flat indoor walkway. The robot was operated in zero-torque mode, compensating only for its own weight and internal joint friction without providing active assistance.

To assess the accuracy of the proposed algorithm, participants walked through a 5-meter segment for each trial, instrumented with a commercial optical gait analysis system (OptoGait, Microgate, Italy), which served as the reference for spatiotemporal gait parameters. These parameters included walking speed, step length, stride length, cadence, and swing phase ratio. During each trial, internal signals from the robot were recorded and later post-processed to extract spatiotemporal gait parameters. These were then compared against the summary outputs provided by the OptoGait system.

It is noted that the 5-meter walkway was used solely for the purpose of validating the algorithm against the OptoGait reference system. For practical deployment, the proposed gait analysis algorithm does not require a predefined walkway length and can be applied during unconstrained overground walking, provided the user performs at least a few consecutive strides to ensure robust estimation (e.g., more than 10 strides, including initial acceleration from a stationary position).

B. DATA PROCESSING AND NORMALIZATION

Internal signals, including inertial measurements from embedded IMUs and hip joint angles from encoders, were continuously recorded during each trial. Spatiotemporal and kinematic parameters were estimated in real time using the embedded algorithm, which operated at a control frequency of 1 kHz. Gait cycles were segmented using the foot contact detection method described in Section III-A.

Joint angle signals were filtered using a zero-phase low-pass Butterworth filter with a cutoff frequency of 20 Hz, which removed high-frequency noise while preserving the original timing characteristics. The resulting data were

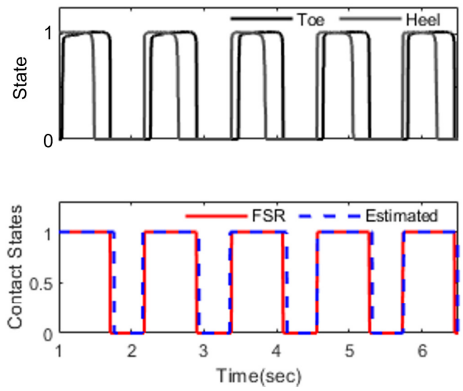

FIGURE 7. Foot contact comparison during representative straight-line walking trials. (a) Binary foot contact states (1: contact, 0: no contact) from heel and toe FSR sensors. (b) Foot contact states derived from FSR sensor thresholds (red) and estimated by the proposed algorithm (blue dashed line) are overlaid to illustrate temporal alignment. For clarity, only selected representative gait cycles are shown.

TABLE 2. Mean absolute errors(MAE), standard deviations(SD), and mean absolute percentage errors(MAPE) of estimated parameters compared to the reference system (OptoGait).

Parameter (Unit)	MAE±SD	MAPE (%)
Swing phase ratio (%)	2.27 ± 2.44	6.17
Cadence (steps/min)	1.97 ± 3.21	1.85
Walking speed (m/s)	0.09 ± 0.12	7.13
Step length (m)	0.04 ± 0.05	5.96
Stride length (m)	0.08 ± 0.10	5.94

resampled onto a normalized gait cycle (0–100%) through linear interpolation, allowing for consistent comparison and averaging across gait cycles within individual subjects. Furthermore, to ensure the assumption of data independence for statistical analysis and to account for the unequal number of trials across participants, the metrics obtained from all successful trials for each participant were averaged, resulting in a single representative value per subject(N=17) for subsequent ICC and linear regression computations.

V. RESULTS

A total of 64 walking trials were conducted using the SUIT H10 robot across 17 healthy adult participants. The proposed embedded gait analysis system estimated spatiotemporal and kinematic parameters in real time. These estimates were validated against a commercial optical gait analysis system.

A. GAIT EVENT DETECTION

Time-series outputs from the real-time foot contact estimation are presented in Fig. 7. To provide a preliminary verification of the intermediate step of foot contact state detection described in Section III-A, an independent assessment was conducted using insole-type Force-Sensitive Resistor (FSR) sensors placed at the heel and toe regions. Three healthy adult subjects participated in walking trials conducted on a 30-meter straight path at their self-selected comfortable

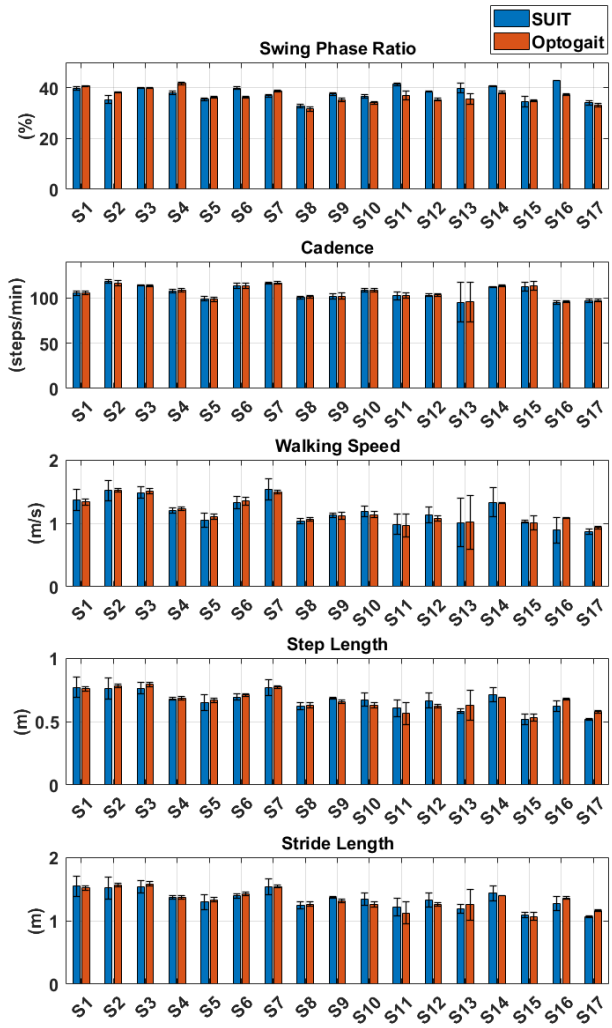

FIGURE 8. Comparison of spatiotemporal gait parameters estimated by SUIT H10 (blue) and OptoGait system (red) for each participant (S1–S17). Data are presented as mean ± SD.

speeds. Results demonstrated high temporal agreement between the estimated and FSR-measured foot contact events, with mean timing differences of -9 ± 48 ms for heel strike events and -39 ± 47 ms for toe off events. Although the verification was conducted with a small number of participants, the observed timing errors remained within 50 ms, which has been reported as acceptable for real-time gait event detection in wearable systems [23], [24].

B. SPATIOTEMPORAL PARAMETER ESTIMATION

The system's spatiotemporal parameter estimation was evaluated against a commercial reference system using foot contact detection. Five key spatiotemporal parameters were compared across all trials, and the results are summarized in Fig.8. For the correlation and consistency analyses presented in Fig.9, a statistical framework based on subject-averaged data was employed to ensure that the unequal contribution of trials across participants did not violate the assumption of data independence. The mean absolute percentage

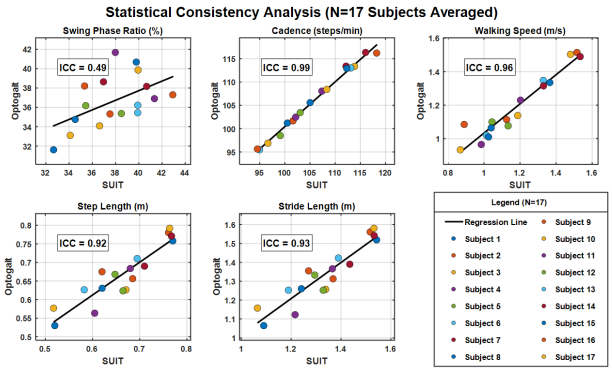

FIGURE 9. Intraclass correlation coefficient (ICC) and regression of data from SUIT and OptoGait for each participant were plotted.

error (MAPE) values for swing phase ratio, cadence, and step/stride lengths were all below 6.2%, indicating strong agreement with the reference system (Table 2). Walking speed exhibited slightly higher percentage error but showed the highest correlation (Fig.9), suggesting that adjusting the scale factor may improve accuracy.

VI. DISCUSSION

The proposed embedded gait analysis algorithm showed strong concordance with reference data, successfully estimating key spatiotemporal and kinematic parameters using only onboard sensors from a hip-assistive wearable robot. Despite relying solely on hip joint encoders and inertial signals, the system achieved agreement with a commercial reference system within a 6.2% error margin for all parameters except walking speed. Intraclass correlation coefficients (ICCs) were also used to quantify the consistency between estimated gait parameters and reference values across participants. Walking speed showed relatively higher absolute and percentage errors, likely due to the combined influence of two steps of estimation, but still exhibited a strong correlation with the reference (ICC = 0.96), suggesting that parameter scaling and individual calibration could improve its precision.

Cadence exhibited the highest agreement with the reference system (ICC = 0.99, MAPE = 1.85%). Step length and stride length also showed moderate to good agreement (ICC = 0.92 and 0.93), while the swing phase ratio yielded relatively lower reliability (ICC = 0.49). This lower consistency in swing phase estimation is primarily attributed to the inherent limitation of the sensor configuration; unlike cadence or speed, which are derived from repetitive kinematic patterns, the swing phase ratio is highly sensitive to the precise timing of foot-off and foot-contact events. Since the SUIT H10 relies solely on hip joint encoders without distal sensors (e.g., foot pressure insoles or ankle IMUs), capturing the exact moments of toe-off and heel-strike remains challenging. Nevertheless, the estimation errors observed in this study fall within acceptable ranges reported in prior research on wearable gait assessment [25], further validating the potential of onboard gait analysis in real-world applications.

A. CLINICAL AND ASSISTIVE APPLICATIONS

The ability to estimate gait parameters without additional sensors or laboratory equipment makes this system suitable for clinical gait assessment. During short walking sessions, the SUIT H10 robot transmits real-time gait estimates to a paired tablet application, allowing live visualization of gait events and hip joint kinematics. This enables clinicians to identify irregular gait patterns and adjust assistance parameters in real time based on observed performance. Furthermore, the consistent post-processing of spatiotemporal metrics supports longitudinal monitoring of rehabilitation progress, helping track treatment outcomes over time.

In wearable assistive robot research, quantifying the effects of robotic assistance remains a challenging task. The proposed algorithm provides a practical solution for evaluating assistance quality in situ by extracting interpretable gait metrics using the robot's onboard sensors. This facilitates iterative tuning of control strategies and may also serve as a foundation for future adaptive systems.

B. FUTURE PERSPECTIVES AND LIMITATIONS

Although this study focused on offline validation using healthy adults, the embedded algorithm has potential for broader applications. For example, real-time gait tracking could support home-based mobility monitoring in older adults or individuals with early-stage neurological impairments. Similarly, the system may offer feedback for athletes to analyze stride patterns and, with additional processing, potentially assess gait symmetry, although such metrics were not evaluated in this study.

In addition, real-time gait estimation may enable adaptive assistance strategies in future wearable robots. For example, feedback from estimated gait parameters could be used to adjust assistive torque profiles in response to user performance, forming the basis of a human-in-the-loop optimization (HILO) approach. While such closed-loop control was not implemented in the present study, the availability of onboard gait metrics represents a key enabler for future developments.

This study has several limitations. First, the participant pool consisted solely of healthy young adults, which may limit the generalizability of the findings to populations with pathological gait. Second, the validation was conducted only during level-ground walking. This was an intentional design choice to first establish the baseline accuracy of the algorithm under standardized conditions using a gold-standard optical system. Applying the current model to more complex scenarios, such as turning or walking on slopes, requires separate consideration, as these tasks involve fundamentally different biomechanics. For instance, turning induces significant kinematic and kinetic asymmetries between the inner and outer limbs [26], while sloped walking substantially alters spatiotemporal and kinematic parameters compared to level walking [27]. Finally, estimation accuracy may have been affected by simplified modeling assumptions for distal limb segments.

Future work should therefore involve more diverse participant groups and test the algorithm's robustness under various walking conditions, potentially exploring personalized modeling or learning-based approaches to improve performance and generalizability.

VII. CONCLUSION

This study proposed a real-time gait analysis algorithm embedded within a hip-assistive wearable robot, capable of estimating key spatiotemporal and kinematic parameters using only onboard sensors. The system was validated against a commercial optical reference, demonstrating consistent agreement in cadence, stride length, and other core gait metrics across healthy adult trials. Notably, the algorithm operates without external cameras, pressure insoles, or additional wearable sensors, making it well-suited for mobile, in-clinic, or community-based applications.

By enabling quantitative gait assessment through a fully embedded and wireless platform, this approach offers a foundation for real-time monitoring, performance evaluation, and adaptive assistance in wearable robotics. Future work will extend its use to broader populations and investigate integration with closed-loop control strategies for personalized rehabilitation.

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